$$\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{vmatrix} u_{31} & u_{32} & u_{33} \end{vmatrix} \begin{pmatrix} x_3 \end{pmatrix} = \begin{pmatrix} u_{13}x_1 + u_{23}x_2 + u_{33}x_3 \end{pmatrix} \begin{pmatrix} x_3 \end{pmatrix}$$

$$(u_{11}x_1 + u_{21}x_2 + u_{31}x_3)x_1 + (u_{12}x_1 + u_{22}x_2 + u_{32}x_3)x_2 + (u_{13}x_1 + u_{23}x_2 + u_{33}x_3)x_3 =$$

$$u_{11}x_1 + u_{21}x_1x_2 + u_{31}x_1x_3 + u_{12}x_1x_2 + u_{22}x_2 + u_{32}x_2x_3 + u_{13}x_1x_3 + u_{23}x_2x_3 + u_{33}x_3$$

$a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + (a_{21}+a_{12})x_1x_2 + (a_{31}+a_{13})x_1x_3 + (a_{32}+a_{23})x_2x_3 \Rightarrow$ quadratic form

$u_{11} \rightarrow$ Coefficient of x_1

$u_{22} \rightarrow$ Coefficient of x_2

$u_{33} \rightarrow$ Coefficient of x_3

Three Variables Example

$u_{21}+u_{12} \Rightarrow a_{12} = a_{21} = \frac{1}{2}$ coefficient of x_1x_2

$u_{31}+u_{13} \Rightarrow a_{13} = a_{31} = \frac{1}{2}$ coefficient of x_1x_3

$u_{32}+u_{23} \Rightarrow a_{32} = a_{23} = \frac{1}{2}$ coefficient of x_2x_3

$x_1 \quad x_2 \quad x_3$

x_1	u_{11}	u_{12}	
x_2	u_{21}	u_{22}	
x_3	u_{31}	u_{32}	

coef of x_1	0.5 coef of x_1x_3	0.5 coef of x_1x_2
0.5 coef of x_1x_2	coef of x_2	0.5 coef of x_2x_3
0.5 coef of x_1x_3	0.5 coef of x_2x_3	coef of x_3